

SkillSpark

Project Team

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Chapter 1

Introduction

The purpose of our app skill spark is that it provides a platform for autistic children that improves their learning and emotional control through gamification. It is designed in a way that is personalized according to specific needs and obsessions of autistic children, making learning more effective and engaging.

Individuals suffering from autism face a lot of challenges in traditional learning environments as they are prone to sensory overload, rigid structures and content that they find unengaging. Thus, SkillSpark , with the help of AI driven technologies creates a flexible system that utilizes real time video feedback and emotional analysis to give kids the access to an entertaining, sensory-friendly learning environment.

1.1 Problem Statement

A recent study found out that 1 in a hundred children suffer from autism. These children lack in things like social interaction, emotional regulation, sensory processing etc. and at the same time excel in different areas e.g problem solving, pattern recognition. These children face difficulties in adjusting to traditional teaching and learning techniques that limit their progress in key developmental areas. They need flexible teaching and learning strategies that accommodate the unique learning styles of each child so that they can reach their full potential.

To solve these issues, we are designing SkillSpark, which provides a gamified learning environment that is customized according to each kid with autism. With the use of dynamic learning exercises based on their unique interests and obsessions , SkillSpark is intended to keep children engaged and motivated. Our solution uses AI-driven technology, real-time feedback, and sensory-friendly designs that adjust to the child's likes and obsessions to overcome the issue of rigid, unengaging learning settings. Through its ability to bridge the gap between traditional education methods and the needs of autistic children, SkillSpark offers a solution that supports the development of both academic progress and emotional health.

1.2 Scope

Write down the scope of your project in a paragraph. Briefly define the main functionalities of the proposed project. Scope defines the boundaries and range of the proposed solution, that would be the part of your project. Write down in logical flow with consistency. (Usually in 14-18 sentences)

1.3 Modules

The goal of this project is to create an interactive, individualized learning platform for children with autism that focuses on strengthening their emotional and cognitive skills through mini-games. The solution would provide real-time personalization of gaming surroundings depending on each child's behaviors and emotional responses, powered by AI. The platform will have dynamic difficulty adjustment, which ensures that the information conforms to the child's learning rate without overstimulation. Key features include games that improve vocabulary, problem-solving, and emotional intelligence, such as a virtual pet game that teaches empathy and problem-solving through daily duties, as well as a vehicle race game that combines math ideas and social safety teachings. The project will also include a caregiver dashboard, which will provide insights into the child's progress, detect triggers using video analysis, and make ideas for game customization to optimize learning results. The overarching goal is to provide a tailored, autism-friendly learning experience that meets individual requirements, fosters emotional growth, and assists caregivers in better understanding the child's learning difficulties.

1.3.1 Skill Spark Mobile App

1. An innovative mobile application specifically designed for autistic children and their caregivers.
2. Integrated games for autistic children
3. Caregivers dashboard for caregivers.
4. Autism friendly UI.

1.3.2 Virtual Pet Game

1. A virtual pet game specifically designed for autistic children. It is a source of learning and entertainment for autistic children with specific animal obsession.
2. MLP based animal choosing
3. MLP based changing of UI/UX.
4. Incorporated Maths learning.
5. Incorporated Social and Safety Skills.
6. Incorporated Alphabet learning.
7. Autism-friendly game concepts.

1.3.3 Road Race Game

1. A car game catering to the obsession of autistic children who love cars. It utilizes that obsession to help them learn things
2. MLP based car choosing.
3. MLP based changing of UI/UX.
4. Road sense learning.
5. Traffic rules learning.
6. Incorporated Social and Safety skills.
7. Incorporated Math and Alphabets learning.
8. Autism-friendly game concepts.

1.3.3 Caregivers Dashboard

1. A caregiver's dashboard to monitor the child's progress and any potential triggers,
2. Monitor the child's game progress.
3. Identify any potential triggers through their emotional analysis.

4. View overall analytics.
5. Modify in-game UI/UX.

1.4 User Classes and Characteristics

User class	Description
Caregiver	A caregiver will set the preferences for the child according to his likes and dislikes and can monitor child's activity and behaviour based on the progress from the caregiver dashboard
Child	The Autistic Children will play integrated games based on their preferences which will help them learn social and education skills while being in their comfort zone .

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 User Stories

The following are several user stories for different modules and features of the application, covering aspects like emotional growth, learning , caregiver interactions etc.

User story 1: Integrated games
As an autistic child, I want to have access to interactive games So that I can learn in a way that is customized to my needs and interests.
Acceptance criteria:
The games adjust their contents based on the child’s interactions.

User story 2: Autism-friendly user interface
As an autistic child, I want the application’s interface to be sensory-friendly So that I am not overstimulated by the bright colors and loud noises while playing the game.
Acceptance criteria:
The user interface has muted colors and calm animations that do not frustrate the child.

User story 3: Caregiver Dashboard

As an caregiver,
I want to keep a check on my childs progress and monitor their emotional and educational growth
So that I can understand their learning preferences and provide additional support accordingly..

Acceptance criteria:

The caregiver can log in to the dashboard and see detailed insights into the child's progress.

User story 4: Customized game characters

As an autistic child,
I want to be given my favourite character according to my caregivers preferences
So that I don't get overwhelmed by choosing it myself.

Acceptance criteria:

The child is presented with a pet of their own liking.

User story 5: Personalized learning

As an autistic child,
I want to learn social and academic skills
So that I can learn while playing with my favourite pet/car.

Acceptance criteria:

The game has interactive sessions on social cues,alphabet and math learning.

User story 6: Emotional learning

As an autistic child,
I want the virtual pet to teach me how to manage my emotions by encouraging me and giving rewards at the end of tasks
So that I can benefit from this and understand how to control my emotions in real life.

Acceptance criteria:

The pet reacts positively by encouraging them even when they fail the tasks given to them.

User story 7: Learning road sense

As an autistic child,
I want the car game to teach me road safety rules

So that I can learn from this and apply them in real life.	
The game includes road signs, traffic lights rules and other safety lessons as part of the gameplay.	

User story 8: Monitor emotional triggers
As a caregiver, I want to identify what triggers my child's emotional stress while they're playing the game. So that I can avoid these triggers in the future.
Acceptance criteria:
Any emotional triggers(e.g frustration, anxiety) that the child experiences while playing are highlighted on the dashboard. .

User story 9: Modifications by caregiver
As a caregiver, I want to change the UI of the game or change its complexity level. So that the game aligned with the child's current needs.
Acceptance criteria:
The caregiver can easily access and modify the UI/UX settings from the dashboard.

User story 10: Educational elements
As an autistic child , I want to integrate math and alphabet learning skills in the game So that I can learn academic skills while playing.
Acceptance criteria:
The game has educational questions integrated in games (e.g solving math problems, recognising alphabets etc).

Use Cases

Use case	Interactive game
Actors	Child
Type	Primary
Precondition	Child has opened the app.
Description	As the game starts, the child selects a game (virtual pet or car game) and starts playing it.

Post condition	The game adapts according to the child's interests and obsessions.
Alternate flow	If the child is not interested in the current game, they can select the other option.

Use case	Customized virtual pet
Actors	Caregiver
Type	Primary
Precondition	Child is playing the virtual pet game.
Description	The caregiver enters the preferences of the child initially and the pet is customized according to these preferences.
Post condition	The preferred virtual pet appears in the game.

Use case	Modify settings
Actors	Caregiver
Type	Primary
Precondition	Caregiver dashboard is opened.
Description	The caregiver dashboard allows them to change difficulty level of tasks, change the user interface to reduce sensory overload, customize the games for (e.g choosing the child's favourite pet as their virtual pet).
Post condition	The changes made by the caregiver are applied to the game in real time.

Use case	Monitor progress
Actors	Caregiver
Type	Primary
Precondition	Caregiver dashboard is opened.
Description	The caregiver dashboard gives a detailed review of the child's learning progress, response times, emotional triggers etc.
Post condition	Caregiver can take actions regarding the feedback received(e.g lowering difficulty level).

Use case	Learning social skills
Actors	Child
Type	Primary
Precondition	Child is playing virtual pet game.
Description	The virtual pet interacts with the child and teaches them different kinds of social skills. It also reacts to the child's actions, helping them understand and practice social skills.
Post condition	The child earns rewards in the form of scores for successfully completing tasks.

Use case	Learning road safety
Actors	Child
Type	Primary
Precondition	The child is playing road race game.
Description	The road race game teaches the child about important road safety rule. It also provides feedback to the child if they are about to make mistakes(e.g getting too close to another car on the road displays a be careful message for the child).
Post condition	The child earns points for following safety rules.
Alternate flow	If the child is about to break a rule, game provides feedback (e.g when the car comes too close to another car on the road, a be careful message is displayed).

Use case	Emotional feedback
Actors	System, child
Type	Supporting
Precondition	Child is playing the game.
Description	The system monitors the child's interactions through response times and emotional analysis.
Post condition	The system provides feedback by adjusting complexity levels to keep the child engaged.

Use case	Modifications by system
Actors	System(AI)
Type	Supporting
Precondition	The child is playing the game.
Description	The system continuously tracks the child's performance. Based on their response times and emotional engagement, it modifies the UI and the complexity levels in real time.
Post condition	The game becomes easier or harder based on how the child is performing.

Use case	Learning academic skills
Actors	Child
Type	Primary
Precondition	Child is playing the game.
Description	Both the games have integrated alphabet and math learning problems so that the child can learn in an environment which does not over stimulate them and keeps them engaged.
Post condition	The child completes tasks and moves to the next level.

2.2 Functional Requirements

2.2.1 Module 1 :Skills spark mobile app

Following are the requirements for module 1:

1. The system shall allow the child to choose between the two games integrated in the app.
2. The system shall customize the game's content based on what the caregiver prefers for the child.
3. The system shall provide an interactive environment for the child during gameplay, including rewards, encouraging comments etc.
4. The system shall analyze the child's interactions to make modifications in real time to reduce the sensory load on the child.
5. The system shall provide default sensory-friendly settings that can be customized by the caregiver or child.

6. The system shall use scenarios within the game to teach the child how to respond to different social situations
7. The system shall analyze the emotions of the child by monitoring through video and audio.

2.2.2 Module 2: Virtual pet game

Following are the requirements for module 2:

1. The system shall allow the caregiver to select a virtual pet according to the caregivers preference.
2. The system shall provide tasks for the child to complete with their virtual pet(e.g collecting grass).
3. The system shall give positive feedback when the child chooses the correct option.
4. The system shall adjust the complexity of the tasks and academic problems based on the reaction time of the child.
5. The system shall integrate alphabet learning , where the child must identify letters or words.

2.2.3 Module 3: Road race game

Following are the requirements for module 3:

1. The system shall integrate math problems into the game.
2. The system shall randomly select a car or select a car based on the caregivers preferences and change it based on the emotion analysis of the child.
3. The system shall include road safety tasks in the game (e.g stopping at red lights).
4. The system shall provide immediate feedback on the child's answers(correct or not) during the race.
5. The system shall monitor the child's performance in road safety tasks and provide feedback.
6. The system shall make changes to the car and change it in real time based on the child's emotions.

2.2.4 Module 4: Caregiver's dashboard

Following are the requirements for module 4:

1. The system shall allow caregivers to monitor their child's progress through the care- giver dashboard.
2. The system shall allow the caregiver to enter the preferences of the child at the start of the game so that it can be customized before the child plays it.

-
3. The system shall display details about the child's performance in games, their response times, completed tasks etc.
 4. The system shall analyze and identify any emotional triggers detected during the game.
 5. The system shall allow caregivers to adjust the game's complexity level through the dashboard.
 6. The system shall allow the caregiver to make changes in the user interface of the game to reduce sensory overload.
 7. The system shall apply these caregiver modified settings in real time.

2.3 Non-Functional Requirements

The following are the nonfunctional requirements of the system. These quality requirements should be specific, quantitative, and verifiable.

2.3.1 Restoration of game

The system must guarantee that upon reloading the game, the restored state accurately reflects the last saved progress of the player by state saving mechanism and data syncing.

2.3.2 Usability

The system shall provide a user friendly interface that is easy to navigate, both for the child and the caregiver. It should have clear instructions with minimum helping screens.

2.3.3 Performance

The system shall load all games within 15 seconds on devices that have minimum 4GB RAM and should respond to user inputs within seconds.

2.3.4 Security

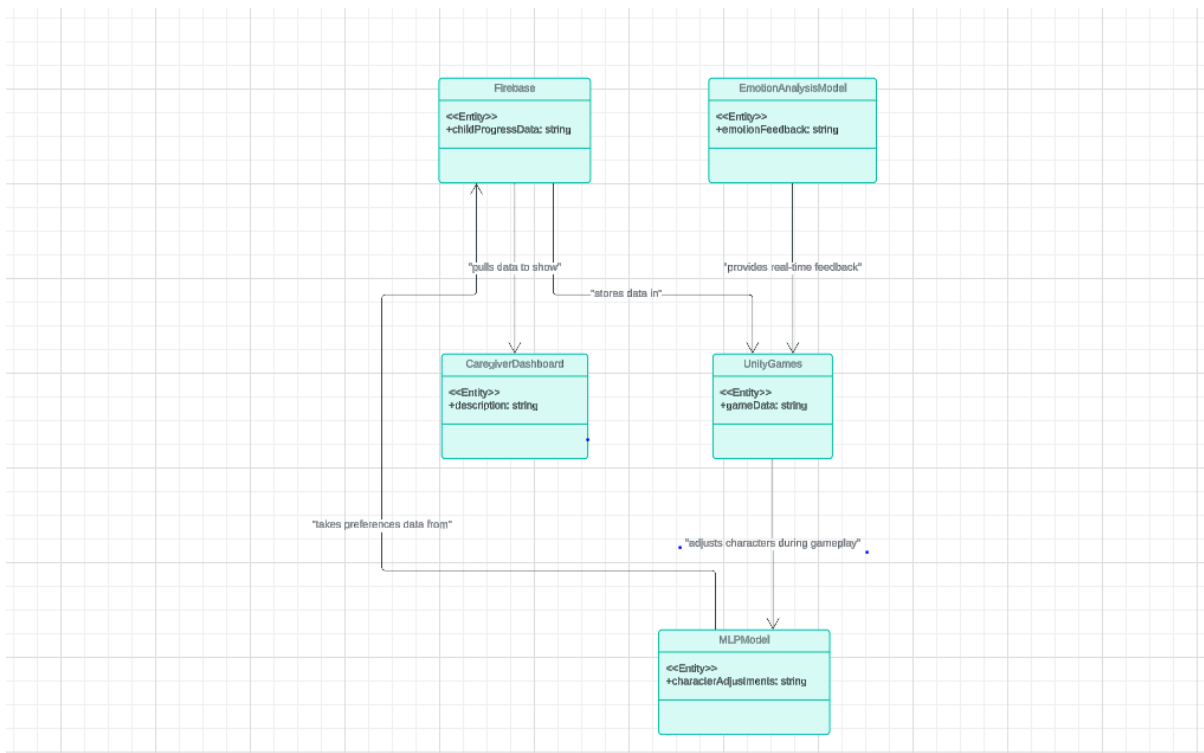
The system shall ensure that the real-time videos are not being stored in any database to prevent unauthorized access. It should notify the caregiver and make immediate modifications to the game without storing sensitive information.

2.3.5 Scalability

The game should be able to run on all kinds of android devices (android smartphones, tablets etc) with adjustable graphic settings. The game should handle 500 concurrent users without any problems.

2.4 Domain Model

Create a representation of the domain model for your project.



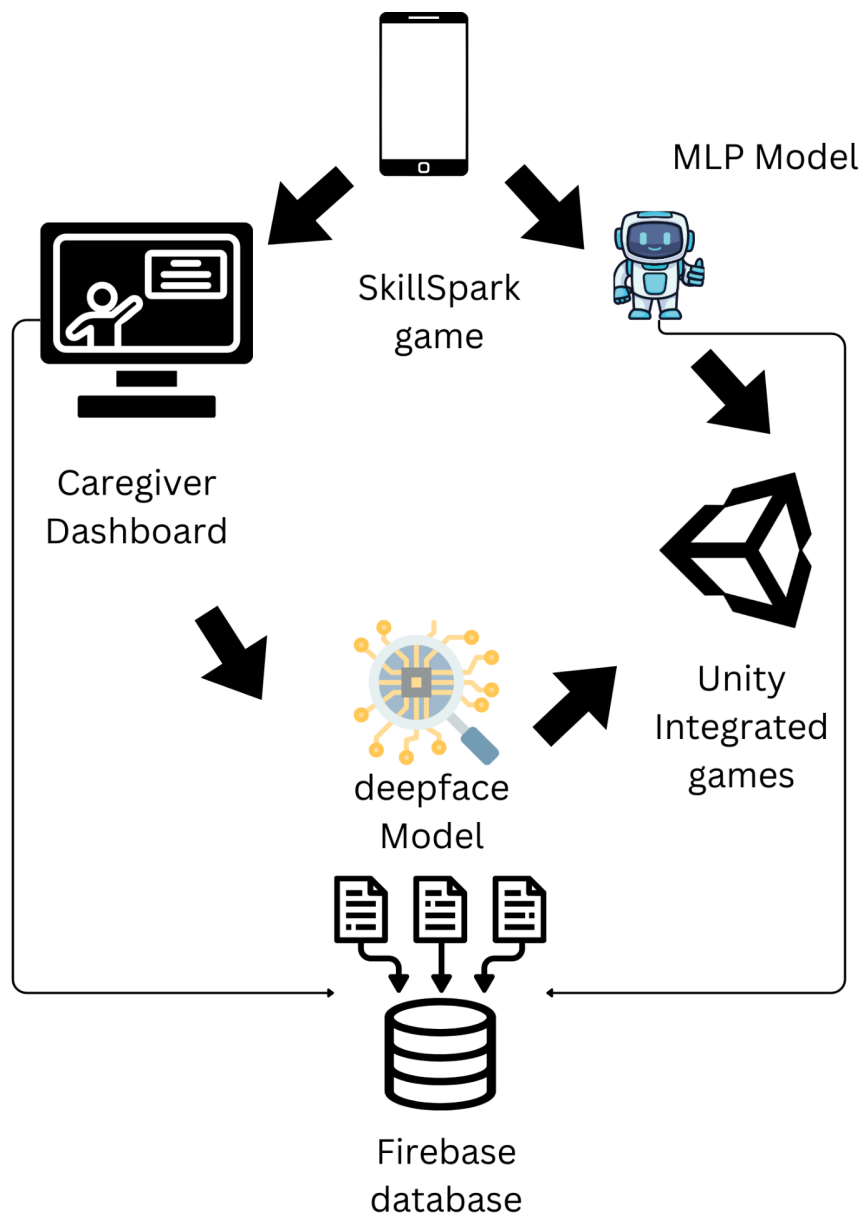
Chapter 3

System Overview

With interactive activities and a caregiver dashboard, this project offers a mobile app for kids with developmental problems (such Down syndrome or autism). The Unity-based games dynamically modify gameplay based on children's behavior analysis with MLP models trained with PyTorch and Keras.

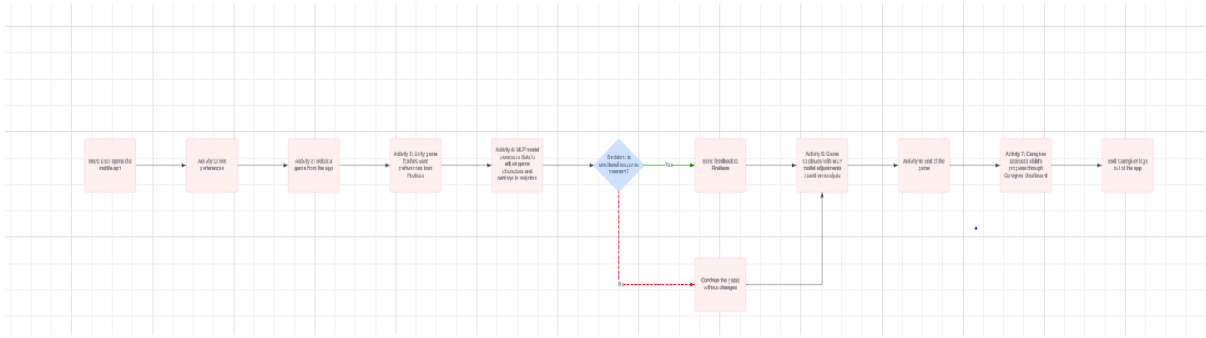
The caregiver dashboard, which was created using React Native, stores all of the data in Firebase and monitors the child's development, feelings, and preferences. A different machine learning algorithm assesses the child's feelings while they play to customize the experience and provide caregivers with information.

Architectural Design

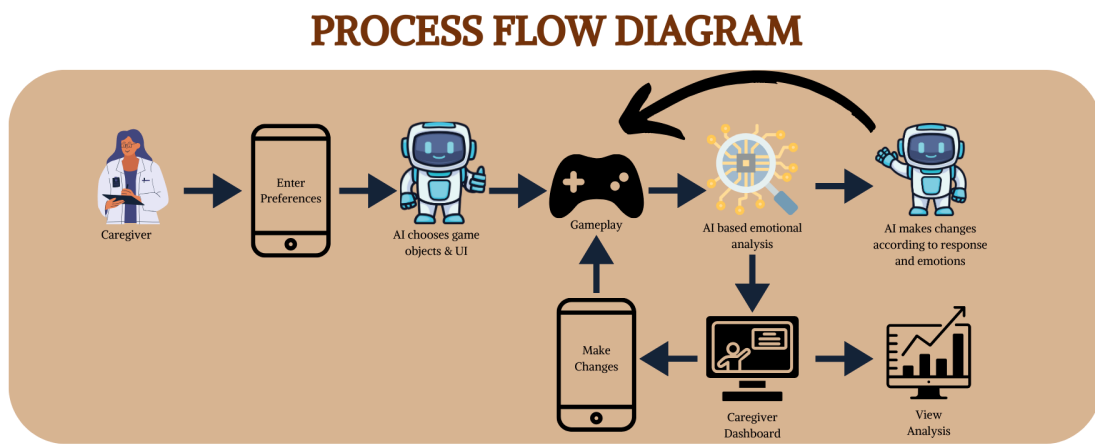


Design Model for Procedural Approach

Activity Diagram



Data Flow Diagram :



System Level sequence Diagram

```
stateDiagram-v2
    [*] --> Idle
    Idle --> CaregiverLogin
    CaregiverLogin --> Dashboard : Successful login
    Dashboard --> SettingPreferences : Caregiver selects "Set Preferences"
    SettingPreferences --> PreferencesSaved : Preferences set
    PreferencesSaved --> GameInProgress : Child starts game
    GameInProgress --> EmotionAnalysisActive : Emotion detected
    GameInProgress --> EmotionAnalysisActive : Emotion analyzed
    GameInProgress --> MLPModelActive : MLP model adjusts character
    GameInProgress --> MLPModelActive : Character adjusted
    GameInProgress --> GameEnded : Game ends
    EmotionAnalysisActive --> GameInProgress
    MLPModelActive --> GameInProgress
    GameEnded --> GameResultsAvailable : Results logged
    GameResultsAvailable --> ProgressViewed : Caregiver checks progress
    ProgressViewed --> Idle : Caregiver logs out
```

Data Design

JSON SCHEMA :

```
{
  "users": {
    "caregiver_id_1": {
      "email": "caregiver1@example.com",
      "preferences": {
        "game_difficulty": "easy",
        "character_choice": "character_A",
      },
      "login_status": "active",
      "children": {
        "child_id_1": {
          "preferences": {
            "game_difficulty": "easy",
            "character_choice": "character_A"
          },
          "progress_reports": {
            "report_id_1": {
              "game_sessions": ["session_id_1", "session_id_2"],
              "emotion_data_summary": {
                "positive_emotions": 75,
                "negative_emotions": 25
              },
              "performance_score": 88
            }
          },
          "emotion_data": {
            "real_time_emotion": "happy",
            "last_update": "2024-10-04T10:20:00Z"
          }
        }
      }
    },
    "game_data": {
      "session_id_1": {
        "child_id": "child_id_1",
        "game_state": {
          "current_level": 2,
          "character_selected": "character_A",
          "score": 1200,
          "time_played": "00:30:00"
        },
        "game_results": {
          "level_completed": true,
          "total_score": 1500,
          "achievements": ["quick_finish", "no_mistakes"]
        }
      }
    }
  }
}
```

```

    },
    "MLP_adjustments": {
      "character_changed_to": "character_B",
      "difficulty_increased": true
    }
  },
  "emotion_analysis": {
    "child_id_1": {
      "real_time_analysis": {
        "emotion_state": "happy",
        "timestamp": "2024-10-04T10:15:00Z"
      },
      "historical_analysis": {
        "session_id_1": {
          "emotions_over_time": [
            {
              "emotion": "happy",
              "timestamp": "2024-10-04T10:10:00Z"
            },
            {
              "emotion": "neutral",
              "timestamp": "2024-10-04T10:12:00Z"
            }
          ]
        }
      }
    }
  },
  "progress_reports": {
    "child_id_1": {
      "report_id_1": {
        "game_sessions": ["session_id_1", "session_id_2"],
        "emotion_data_summary": {
          "positive_emotions": 75,
          "negative_emotions": 25
        },
        "performance_score": 88
      }
    }
  }
}

```

EXPLANATION :

Users' Collection:

Every caregiver has a unique ID (caregiver_id_1), and the child's information (such as preferences, progress reports, and real-time emotion data) is saved under them. Here are saved the preferences of the caregiver, including the game settings they select for the child.

Gathering Gaming Data:

The session_id keeps track of each gaming session. Game-specific information is kept here, including the status of the game, its outcomes, and modifications performed by the MLP model.

Every session's MLP modifications—such as altering characters or level of difficulty—are recorded.

Gathering Emotion Analysis:

Each child (child_id_1) has a record of their history and real-time emotional tracking.

A complete emotional profile for every gameplay session is made possible by the system's ability to store multiple emotions over time.

Progress Reports Gathering:

Here is a summary of the child's development , containing mentions of gaming sessions, summaries of emotional data, and performance ratings.

From the progress reports that have been saved, caregivers can see a comprehensive overview of the child's experience.

Firebase Firestore users can utilize this structure by creating collections such as users, game_data, emotion_analysis, and progress_reports. Each document in these collections represents a specific caregiver, game session, child, etc. This hierarchical structure will be automatically followed by Firebase for both data retrieval and storage.